

Translated

From: e-maxx.ru

Manacher's Algorithm - Finding all sub-palindromes in $O(N)$

Statement

Given string s with length n . Find all the pairs (i, j) such that substring $s[i \dots j]$ is a palindrome. String t is a palindrome when $t = t_{\text{rev}}$ (t_{rev} is a reversed string for t).

More precise statement

In the worst case string might have up to $O(n^2)$ palindromic substrings, and at the first glance it seems that there is no linear algorithm for this problem.

But the information about the palindromes can be kept **in a compact way**: for each position i we will find the number of non-empty palindromes centered at this position.

Palindromes with a common center form a contiguous chain, that is if we have a palindrome of length l centered in i , we also have palindromes of lengths $l-2$, $l-4$ and so on also centered in i . Therefore, we will collect the information about all palindromic substrings in this way.

Palindromes of odd and even lengths are accounted for separately as $d_{\text{odd}}[i]$ and $d_{\text{even}}[i]$. For the palindromes of even length we assume that they're centered in the position i if their two central characters are $s[i]$ and $s[i-1]$.

For instance, string $s = abababc$ has three palindromes with odd length with centers in the position $s[3] = b$, i. e. $d_{\text{odd}}[3] = 3$:

$\overbrace{b \ a \ \underbrace{b}_{s_3} \ a \ b}^{d_{\text{odd}}[3]=3} c$

And string $s = cbaabd$ has two palindromes with even length with centers in the position $s[3] = a$, i. e. $d_{\text{even}}[3] = 2$:

$c \ \overbrace{b \ a \ \underbrace{a}_{s_3} \ b}^{d_{\text{even}}[3]=2} d$